H5. Block structure, a wider dS window, and a codeml check of peak positions

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Follow-up to H4 (`reports/H4_duplication_mode.md`). Three additions: block-level descriptors (retention, strand concordance, copy depth, contig coverage), a wider recent window (dS <= 0.3, neighbours dS <= 0.5), and a codeml second pass for genomes with a possible interior dS peak (Option E). Scripts: `dup_mode.py` (env `DUP_RECENT`, `DUP_NBR`, `DUP_TAG`), `dup_block_tests.py`, `optionE_compare.py`, `option_e_codeml/`.

Status: interim (7,384-genome tables). Mode thresholds are exploratory.

H5.1 Collinear blocks are real syntenic structures

Hypothesis. Gene strands in a block agree with the block orientation (direct blocks: same strand; inverted blocks: opposite strand). Strand is not used to call blocks, so agreement is independent evidence.

It would count against the hypothesis if: strand concordance is near 0.5 (the random expectation).

Test. Per block, share of pairs whose strand relation matches the block orientation; share of blocks with concordance >= 0.9, in informative genomes (window dS <= 0.1). Verdict rule: SUPPORTED if > 85% of blocks have concordance >= 0.9.

Result.	mode	genomes	blocks	median_pairs	strand_concordance	share_strand_concordance_ge_0_9	retention
dispersed_diffuse	182	1.76e+03	3	1	0.887	0.75	2
dispersed_selective	37	80	3.5	1	0.887	0.879	2
mixed	686	3.08e+04	4	1	0.93	0.8	3
multi_contig_blocks	385	6.94e+04	6	1	0.959	0.765	2
single_contig_pair	78	381	5	1	0.958	0.875	2
tandem_array	14	60	4	1	0.833	0.8	2

All modes pooled: 0.949 of blocks have strand concordance >= 0.9. The expectation for unrelated pairs is 0.5 per gene pair.

Verdict: SUPPORTED. This supports the block calls as syntenic runs. It does not say what produced the duplication.

H5.2 Hybrid and doubled-genome blocks are two-copy; repeated segmental duplication is multi-copy

Hypothesis. Genomes with a hybrid or whole-genome-like signal have blocks with copy number 2, while multi-copy blocks (copy number > 2) mark repeated segmental duplication.

It would count against the hypothesis if: named hybrids have as many multi-copy blocks as other block genomes.

Test. Copy number of a block = 1 + the number of other blocks that overlap its busiest region by >= 50%. Share of blocks with copies > 2 in named hybrids against the multi-contig-block mode. Verdict rule: SUPPORTED if the hybrid share is lower.

Result.	genome	blocks	median_pairs	retention	strand_concordance	copies_median	share_multi_copy	share_scaffold_like	coverage_smaller_c
Aspergillus_latus_MOS46169	8	5	0.646	0.833	2	0.25	0	0.00495	0.0154
Cryptococcus_neoformans_x_Cryptococcus_deneoformans_CBS_132	1	3	1	1	2	0	0	0.25	0.00409
Ogataea_polymorpha_x_Ogataea_parapolymorpha_CBS_1977	269	13	1	1	2	0.193	0.543	0.95	0
Puccinia_graminis_21-0	863	6	0.667	1	2	0.301	0.00232	0.0112	0.0345
Verticillium_longisporum_PD589	246	4	0.5	1	2	0.0285	0	0.00654	0.0821
Zygosaccharomyces_bailii_ISA1307	16	3	0.5	1	2	0	0	0.109	0.0764
Zygosaccharomyces_parabailii_ATCC5316	7	3	0.75	1	2	0	0	0.308	0.0776

Mean share of multi-copy blocks: four named hybrids 0.055; multi-contig-block mode 0.424.

Verdict: SUPPORTED. *V. longisporum* and the *Zygosaccharomyces* genomes have 0-3% multi-copy blocks. *Puccinia graminis* 21-0 has about 30%, which fits repeated duplication inside a two-haplotype assembly. Sampled pairs under-count partners in large families, so copy number is a lower bound.

H5.3 Some block genomes are duplicated scaffolds or doubled assemblies

Hypothesis. Blocks that cover >= 80% of the smaller contig with median dS <= 0.01 mark whole duplicated contigs (assembly duplication) and are concentrated in a minority of genomes.

It would count against the hypothesis if: such blocks are spread evenly over all genomes.

Test. Per genome, share of blocks that are scaffold-like. Count genomes with >= 50% scaffold-like blocks.

Result. 35 of 1382 informative genomes with blocks have >= 50% scaffold-like blocks. In the *Ogataea polymorpha x parapolymorpha* hybrid genome 0.54 of 269 blocks are scaffold-like (median block coverage 0.95, median dS 0.000).

Verdict: DESCRIPTIVE. Scaffold-like blocks fit duplicated contigs or a doubled assembly. A real near-identical homeolog pair would look the same. Read depth or contig-level alignment would separate them.

H5.4 A wider window (dS <= 0.3) recovers genome-scale duplication in hybrids whose copies diverged beyond 0.1

Hypothesis. With the wider window, more genomes become informative and hybrids show breadth near 1 instead of a fraction.

It would count against the hypothesis if: the number of informative genomes and hybrid breadth do not change.

Test. Informative genomes (>= 50 eligible recent pairs, null block share <= 0.1) and named-genome breadth and block share for both windows.

Result. Informative genomes: 1,513 (window 0.1) and 2,227 (window 0.3).

genome	breadth_w01	breadth_w03	share_block_w01	share_block_w03	null_w03	mode_w01	mode_w03
Ogataea_polymorpha_x_Ogataea_parapolymorpha_CBS_1977	0.885	0.991	0.991	0.956	0.113	multi_contig_blocks	uninformative
Saccharomyces_pastorianus_CBS_1484	0.00975	0.0367	0	0.0359	0	uninformative	dispersed_diffuse
Zygosaccharomyces_bailii_ISA1307	0.0892	0.949	0.984	0.99	0.0535	multi_contig_blocks	multi_contig_blocks
Zygosaccharomyces_mellis_65-7431	0.19	0.969	0.981	0.993	0.00606	multi_contig_blocks	multi_contig_blocks
Verticillium_longisporum_PD589	0.179	0.828	0.982	0.986	0	multi_contig_blocks	multi_contig_blocks
Aspergillus_latus_MO46169	0.00946	0.0161	0.382	0.331	0	mixed	mixed
Zygosaccharomyces_parabailii_PYCC5316	0.0336	0.613	0.983	0.992	0.602	uninformative	uninformative
Cryptococcus_neoformans_x_Cryptococcus_deneoformans_CBS_132	0.044	0.274	0.244	0.816	0.191	uninformative	uninformative
Zygosaccharomyces_pseudobailii	0.0875	0.879	0.935	0.99	0.00424	multi_contig_blocks	multi_contig_blocks
Puccinia_graminis_21-0	0.655	0.792	0.601	0.505	0.000606	multi_contig_blocks	multi_contig_blocks

Verdict: SUPPORTED. The window ties the interpretation to dS: *V. longisporum* blocks have median dS 0.14 and *Z. bailii* 0.16, both above the first window. *A. latus* stays 'mixed' (block share 0.33). The *Cryptococcus* AD hybrid has a null block share of 0.19 at the wider window and stays unclassed.

H5.5 Genome-scale block genomes form a candidate list that overlaps assembly evidence

Hypothesis. Genomes with breadth >= 0.3, block share >= 0.5 and blocks on >= 5 contig pairs (window 0.3) also show high BUSCO duplication or a sharp interior peak.

It would count against the hypothesis if: BUSCO duplication and peak tiers are no higher in the candidates than in other informative genomes.

Test. Candidate list from `dupmode_genome_scale_candidates_w03.tsv`; BUSCO duplicated-% bins; overlap with shape tiers A and B; genera.

Result. 129 candidate genomes; 77 Basidiomycota, 28 Ascomycota, 16 Microsporidia. Shape tier A or B: 28; any interior peak <= 1.0: 91.

BUSCO dup %	genomes
<10	15
10-40	56
>=40	58

Largest genera:

genus	genomes
Mycena	13
Apiotrichum	6
Amblyosporidae	5
Epichloe	5
Nosema	4

genus	genomes
Lactarius	4
Pisolithus	3
Vairimorpha	3
Armillaria	3
Zygosaccharomyces	3

First 25 by breadth:

genome	class	breadth	share_block	share_block_hk	n_block_contig_pairs	block_median_dS	block_dS_mad	busco_dup	genes_ratio_clade	tier
Amblyospora_sp._gen/OTU018_MHUK	Microsporidia_unassigned	[illegible]	[illegible]	[illegible]	[illegible]	0.0686	0.0194	10.2	6.36	D broad hump
Lentinula_aciculospora_JLM2183	Agaricomycetes	0.962	[illegible]	0.00878	226	0.00114	0.00114	92.2	1.76	
Zygosaccharomyces_mellis_6-7431	Saccharomycetes	0.958	0.993	0.00606	20	0.125	0.0013	64.3	1.89	A sharp + BUSCO dup>=40%
Zygosaccharomyces_mellis_ISA1307	Saccharomycetes	0.949	[illegible]	0.0535	108	0.159	0.00646	64.1	1.97	A sharp + BUSCO dup>=40%
Lodderomyces_metapsilosis_4146	Pichiomycetes	0.948	[illegible]	0.00464	72	0.146	0.0212	70.9	2.23	A sharp + BUSCO dup>=40%
Nosema_sp._gen/OTU0234_Ox001	Microsporidia_unassigned	[illegible]	[illegible]	0.0072	251	0.0385	0.0169	10.3	3.25	D broad hump
Amblyospora_sp._gen/OTU0195_Ox001	Microsporidia_unassigned	[illegible]	[illegible]	[illegible]	256	0.0942	0.0888	10.2	3.95	D broad hump
Apiotrichum_sp._Y4201-1_Y4201-1	Tremellomycetes	[illegible]	0.998	0.0395	146	0.151	0.0176	74.7	1.85	A sharp + BUSCO dup>=40%
[illegible]	Sordariomycetes	[illegible]	[illegible]	0.00485	714	0.0683	0.0194	91.5	1.99	A sharp + BUSCO dup>=40%
Apiotrichum_sp._PDH3504B_PDH3304B	Tremellomycetes	[illegible]	[illegible]	0.0464	170	0.153	0.0171	75	1.86	A sharp + BUSCO dup>=40%

genome	class	breadth	share_block	share_block_bulk	n_bulk_contig_pairs	block_median_dS	block_dS_mad	busco_dup	genes_ratio_clade	tier
Apiotrichum_sp._PA4207_PA4207	Tremellomycetes	0.924	0.998	0.061	177	0.151	0.0175	74.9	1.87	A sharp + BUSCO dup>=40%
Apiotrichum_sp._PL03829B_PL03829B	Tremellomycetes	0.934	0.998	0.0397	177	0.153	0.0171	74.2	1.86	A sharp + BUSCO dup>=40%
Apiotrichum_sp._Y4204_Y4204	Tremellomycetes	0.923	0.997	0.0369	145	0.149	0.0163	74.9	1.86	A sharp + BUSCO dup>=40%
Apiotrichum_sp._PL03826B_PL03826B	Tremellomycetes	0.935	0.997	0.0494	174	0.151	0.0174	74.2	1.86	A sharp + BUSCO dup>=40%
Nosema_[illegible]	Microsporidia/unassigned	0.921	0.789	0.0144	107	0.0184	0.0184	10.2	2.23	
Puccinia_triticina_[illegible]	Pucciniomycetes	0.762	0.505	0.000672	327	0.00919	0.00919	89.3	2.84	
Candida_sanyaensis_SD423	Pichiomycetes	0.918	0.946	0.0289	272	0.0382	0.0119	59.9	2.28	D broad hump
Cryomyces_antarcticus_CBS_116301	Dothideomycetes	0.916	0.967	0.012	178	0.0082	0.00625	87.7	1.72	
Scleroderma_citrinum	Agaricomycetes	0.906	0.812	0.00691	147	0.000276	0.000276	89.9	1.72	
Nosema_sp._[illegible]	Microsporidia/unassigned	[illegible]	0.921	0.00789	206	0.0432	0.0409	8.1	3.51	D broad hump
Epichloe_hybrida_Lp1	Sordariomycetes	0.894	0.967	0.00513	346	0.0868	0.0171	86.3	1.44	A sharp + BUSCO dup>=40%
Zygosaccharomyces_pseudobailii	Saccharomycetes	0.879	0.99	0.00424	25	0.212	0.00832	61.2	1.88	A sharp + BUSCO dup>=40%
Epichloe_sp._1_LKC-2024_1_LKC-2024	Sordariomycetes	0.876	0.986	0.00102	126	0.0996	0.00942	83.3	1.29	A sharp + BUSCO dup>=40%
Epichloe_sp._2_LKC-2024_2_LKC-2024	Sordariomycetes	0.875	0.988	0	143	0.099	0.00911	82.7	1.28	A sharp + BUSCO dup>=40%

genome	class	breadth	share_block	share_block_null	n_block_contig_pairs	block_median_dS	block_dS_mad	busco_dup	genes_ratio_clade	tier
Epichloe_sp_FaTG-3_NFe1100	Sordariomycetes	0.861	0.984	0.00777	256	0.0954	0.0106	76.4	1.26	A sharp + BUSCO dup>=40%

Verdict: DESCRIPTIVE. The list is a screen. Several strains of one genus count as separate genomes (for example the Apiotrichum strains), so the number of independent events is lower than the number of genomes. Microsporidia entries have BUSCO duplication near 10% and gene counts several times the clade median; they need a separate check for contamination or assembly doubling.

H5.6 yn00 peak positions below dS 1.0 are close to codeml positions

Hypothesis. For genomes with an interior peak, the peak position from codeml is within 10% of the yn00 position, so the screens (H1, H4, H5) do not move when the estimator changes.

It would count against the hypothesis if: codeml peak positions differ from yn00 by more than 10% in most genomes.

Test. Same pairs-per-node sampling and gates; codeml run on 127 genomes with a possible peak (shape tiers A to C, excluding 8 genomes that already had codeml). The shape detector is applied to codeml node dS; the position is the median node dS inside the peak window. Verdict rule: SUPPORTED if >= 70% of genomes have a ratio between 0.9 and 1.1.

Result. Peak detected in both estimators: 126 of 127 genomes. Position ratio codeml / yn00: median 1.032, 10th-90th percentile 1.004 to 1.079; 0.93 of genomes within 10%.

bin	n_nodes	median_yn00	median_codeml	median_ratio	p10	p90	share_within_20pct
(0.0, 0.05]	30336	0.0377	0.038	1.02	0.856	1.18	0.894
(0.05, 0.1]	112704	0.0759	0.0777	1.03	0.94	1.11	0.964
(0.1, 0.2]	149289	0.143	0.147	1.03	0.943	1.13	0.95
(0.2, 0.3]	93298	0.244	0.253	1.04	0.937	1.16	0.931
(0.3, 0.5]	92701	0.383	0.396	1.04	0.912	1.2	0.882
(0.5, 0.8]	83696	0.631	0.648	1.03	0.868	1.26	0.811
(0.8, 1.2]	54710	0.951	0.925	0.971	0.771	1.29	0.709
(1.2, 1.6]	23799	1.37	1.27	0.938	0.654	1.41	0.478
(1.6, 2.5]	14910	1.82	1.74	0.936	0.551	1.27	0.465

Verdict: SUPPORTED. This node-level result differs from the earlier wave-1 comparison (300 genomes, `tables/yn00_vs_codeml_nodes_binned.tsv`), which gave codeml / yn00 ratios of 1.08 to 1.13 for yn00 dS 0.2 to 1.6. Here the ratio is about 1.03 below dS 0.8 and 0.94 to 0.97 above 0.8. The genome sets differ (wave-1: all taxa; here: genomes with a possible peak) and the node matching differs. I have not resolved the difference. One untested possibility is regression to the mean, because nodes are binned by the yn00 value and the two runs sample different pairs. What this test does show is that peak positions below dS 1.0 agree within 10% in most genomes. Peaks above dS 1.0 are not assessed. One genome (Hemileccinum hortonii, tier C) has a peak with yn00 and none with codeml.

Follow-ups

- Contig-level alignment or read depth for the scaffold-like genomes and for the candidate list, to separate duplicated assemblies from real hybrids.
- Treat strains of one genus (for example Apiotrichum) as one candidate event.

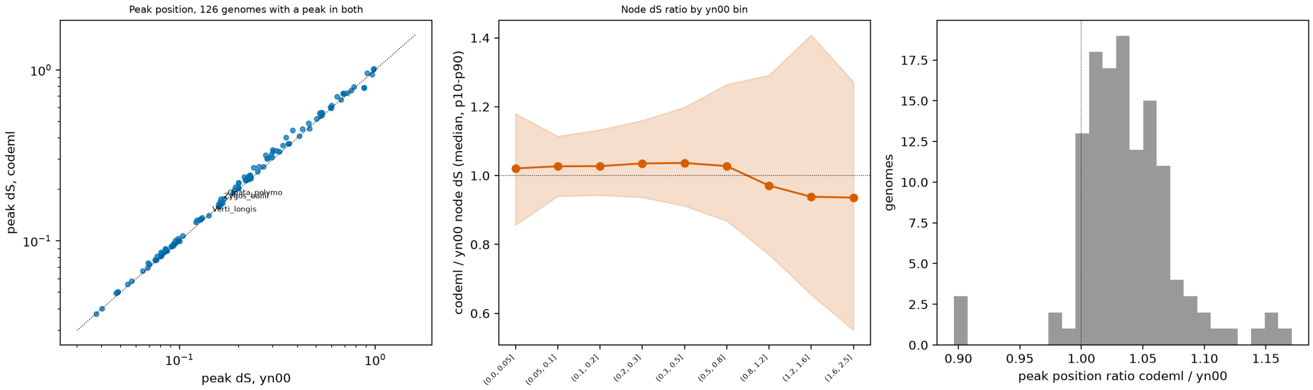

Figure 1: Option E

- Microsporidia candidates: check contamination and assembly doubling.
- Phase the two block partners in V. longisporum, Z. bailii and Epichloe coenophiala against relatives (gene-tree placement) to test for allopolyploidy; gene order and dS cannot do this.
- Rerun on the final tables.

Reproduce

`DUP_TAG= scripts/dup_mode.py`; `DUP_RECENT=0.3 DUP_NBR=0.5 DUP_TAG=_w03 scripts/dup_mode.py`; `scripts/dup_mode_tests.py` and `scripts/dup_block_tests.py` (same env); `option_e_codeml/select_genom`
`submit_array.sh`; `scripts/optionE_compare.py`; `scripts/make_h5_report.py`.